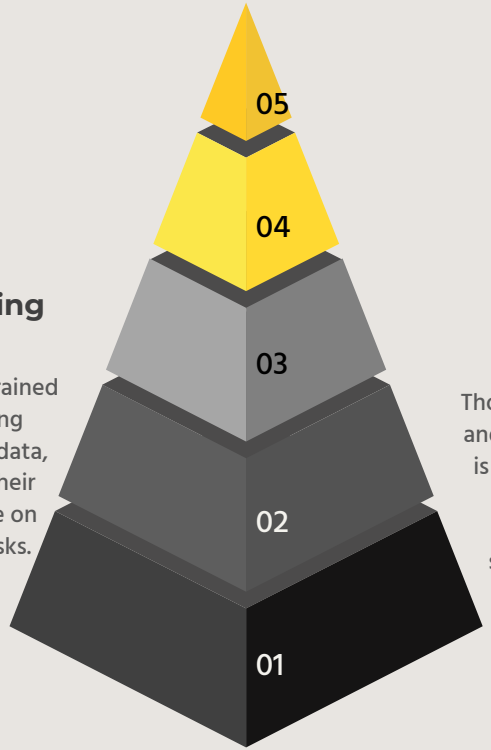
H2O.ai University

LLM Learning Path - Level 1

Author: Andreea Turcu

Head of Global Training @H2O.ai

Building Steps for LLMs



01 Foundation

Powerful language models trained on extensive text data, forming the basis for various language tasks.

02 DataPrep

Converting documents into instruction pairs, like QA pairs, facilitating fine-tuning and tasks.

03 Fine-tuning

Refining pre-trained models using task-specific data, enhancing their performance on targeted tasks.

04 Eval LLMs

Thoroughly assessing and comparing LLMs is increasingly vital due to their heightened significance and complexity.

05 Database

Effectively utilize company data with a database that seamlessly integrates new PDFs, eliminating the need for model retraining.

06 Applications

Elevate interactions with advanced language comprehension and LLM-driven response generation for enriched user experiences.

H2O.ai Confidential

Table of Contents

1. Introduction to Language Models
2. Understanding LLM Architecture / Foundation Models
3. Getting Started with LLM Data Studio
4. Fine-tuning LLMs
5. Making Your Own GPT and Fine-tuning using LLM Studio
6. Evaluating and Benchmarking LLMs
7. Practical Applications and Case Studies

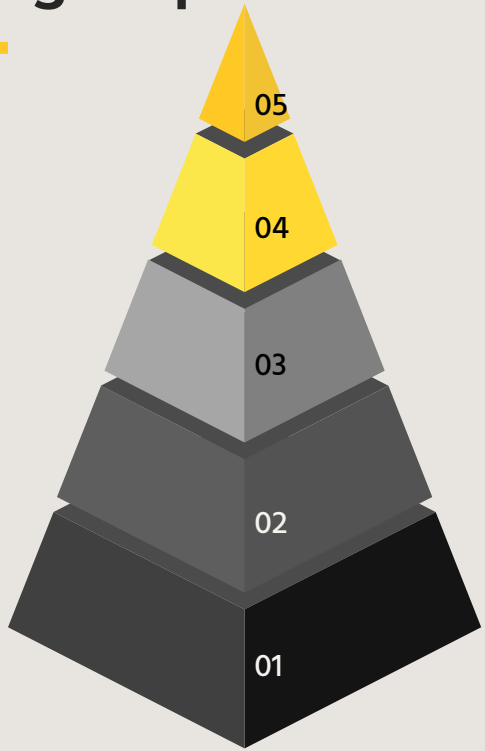
H2O.ai Confidential

Contents at a Glance

1. Introduction to Language Models
    - What is a Language Model?
    - Techniques Commonly Used
    - Importance and Applications

H2O.ai Confidential

Building Steps for LLMs



01

Foundation

Powerful language models trained on extensive text data, forming the basis for various language tasks.

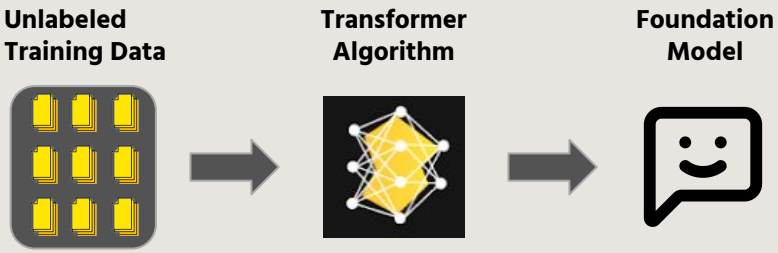
Contents at a Glance

1. Introduction to Language Models
2. Understanding LLM Architecture / Foundation Models
   - What are Foundation Models?
   - Neural Networks and Deep Learning
   - Transformer Architecture vs. LLM Architecture
   - Pre-training & fine-tuning of LLMs
   - Transfer Learning and Adaptation

H2O.ai Confidential

Generative AI Definitions

Foundation Models



Large Language Models (LLMs)

Generative AI
Collection of ML algorithms that learn a representation of artifacts from data and models, and use it to generate brand-new, completely original artifacts that preserve a likeness to original data or models.

Foundation model
Is a Large machine learning model trained on a large amount of unlabeled data using a transformer algorithm. This model can be augmented by a range of fine-tuning (adapter) techniques. The resulting model can be further adapted to a wide range of applications.

Large Language Model (LLM)
An LLM is a type of foundation model specifically designed for natural language processing.

Generative Pre-trained Transformer (GPT)
Is an LLM specifically designed to predict the next token. For example ChatGPT is a conversational application built on top of an LLM.

H2O.ai Confidential

Essential topics:

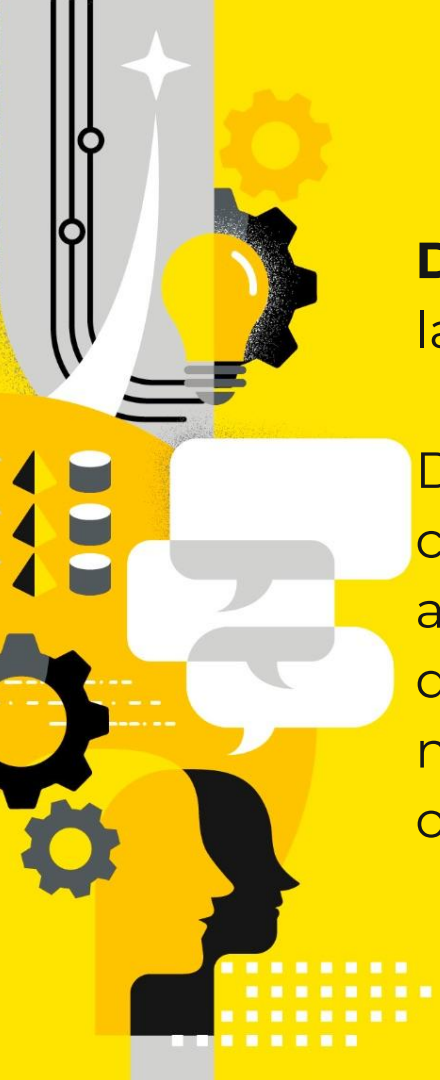
1. Grasping the essence of Foundation Models
2. Delving into Neural Networks and Deep Learning
3. Exploring the intricacies of the Transformer Architecture
4. Understanding the concepts of pre-training and fine-tuning in LLMs
5. Navigating Transfer Learning and Adaptation techniques

Foundation models can be used for a wide range of tasks:

1. Answering questions
2. Generating human-like text
3. Translating languages
4. Creating chatbots
5. Summarizing articles, and more

Neural Networks

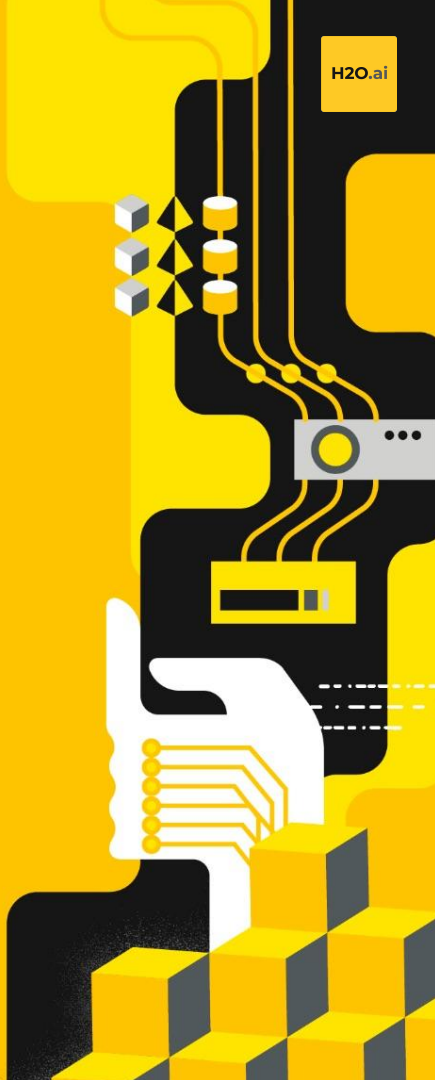
Each node receives input from multiple nodes in the previous layer, performs a computation, and passes the output to the next layer. The output of the last layer represents the final prediction or decision made by the neural network.



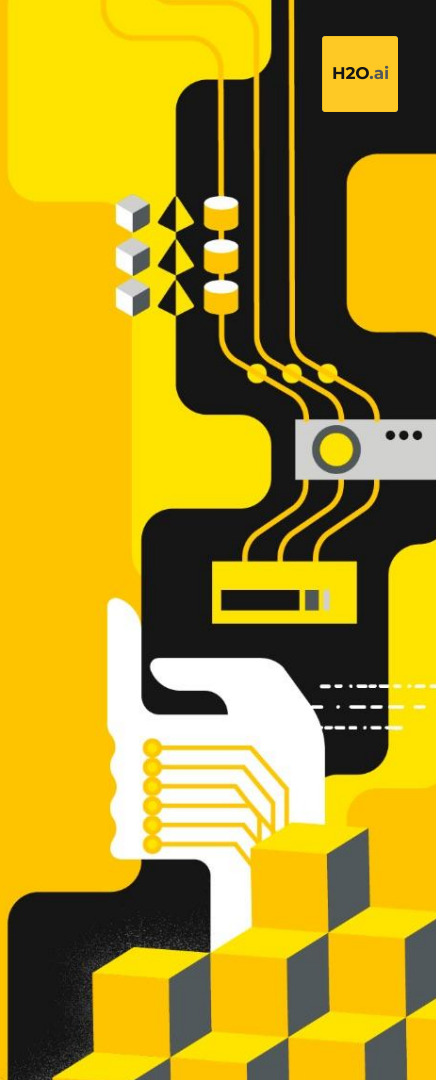
Deep Learning = Neural networks with multiple layers

Deep learning models are capable of learning complex patterns and representations from large amounts of data. The term "deep" refers to the depth of the network, which signifies the number of hidden layers between the input and output layers.

In **forward propagation**, input data flows through the network, transforming into a meaningful output.



Backpropagation fine-tunes network parameters by minimizing prediction errors through iterative adjustments based on desired output.



To remember:

- Not all neural networks qualify as deep learning models.
- Deep learning is distinguished by network depth.
- Depth enables the learning of intricate data features and relationships.
- This leads to improved performance in tasks like image recognition and natural language processing.

Applications of Neural Networks and Deep Learning in LLMs:

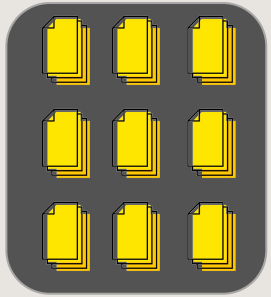
- Natural Language Processing (NLP)
- Speech Recognition
- Recommendation Systems
- Text Generation
- Language Understanding and Context
- Automation and Efficiency
- User Experience Enhancement

- The emergence of Large Language Models (LLMs) coincided with advancements in language understanding and generation.

- LLMs are distinguished by their exceptional size and complexity.
  - These models consist of billions of specialized components.
  - These components enable LLMs to comprehend intricate language nuances.

- LLMs are capable of generating high-quality text.

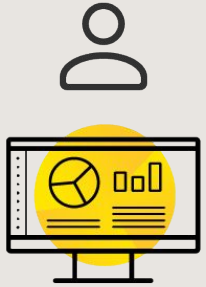
Fine-tune Example:
Learn a Specific Style of Answering and Writing



Foundation
Large Language Model

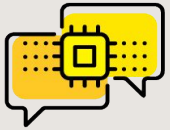
Autoregressive, trained on diverse data (“the whole internet”). Good at continuing text.

Data Scientist



Fine-tuning training
Hyperparameter tuning

Fine-Tuned
Large Language Model



Specialized style: learned prompt & answer, instructions

H2O.ai Confidential

Crucial Role in Language Models

1. Enhanced Communication
2. Information Assessment
3. Ethical Implications
4. Prospects for the Future

H2O.ai Confidential

Key Areas where LMs are used:

1. Chatbots and Virtual Assistants
2. Language Translation
3. Content Generation
4. Sentiment Analysis
5. Text Completion and Auto-correction
6. Voice Assistants

H2O.ai Confidential

Distinguishing Characteristics of LLMs

1. Scale
2. Creative Writing
3. Complex Problem Solving
4. Domain Expertise
5. Enhanced Language Understanding
6. Data Efficiency
7. Pre-training and Fine-tuning
8. Contextual Understanding
9. Language Generation
10. Transfer Learning
11. Versatility and Applications
12. Research and Innovation

H2O.ai Confidential

Some important terms related to the Transformer architecture:

1. Attention
2. Multi-head Attention
3. Encoder
4. Decoder
5. Self-Attention
6. Feed-Forward Neural Network
7. Positional Encoding
8. Masking

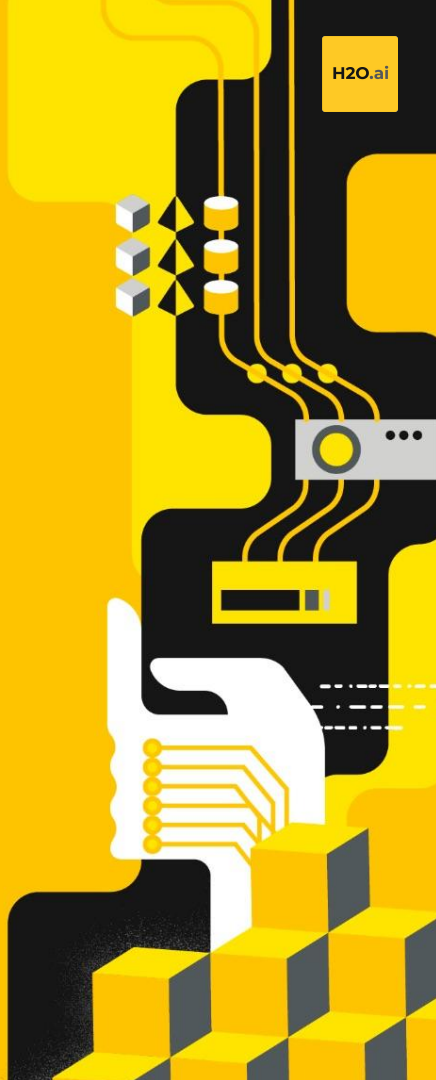
H2O.ai Confidential

Reminder

- The Transformer is a specialized neural network architecture introduced in the research paper "Attention is All You Need."

- Its primary function is to process sequences of data.

- It utilizes self-attention, a distinctive mechanism, to efficiently capture relationships between words within a sentence.



Reminder

- Large Language Models fall within a broader category of models trained on extensive textual data without human annotations.

- Prominent models such as GPT-3 and BERT are constructed based on the underlying Transformer architecture.

- These models attain comprehensive language representations by harnessing the abundant data they encounter during their training process.

Primary objective of LLMs

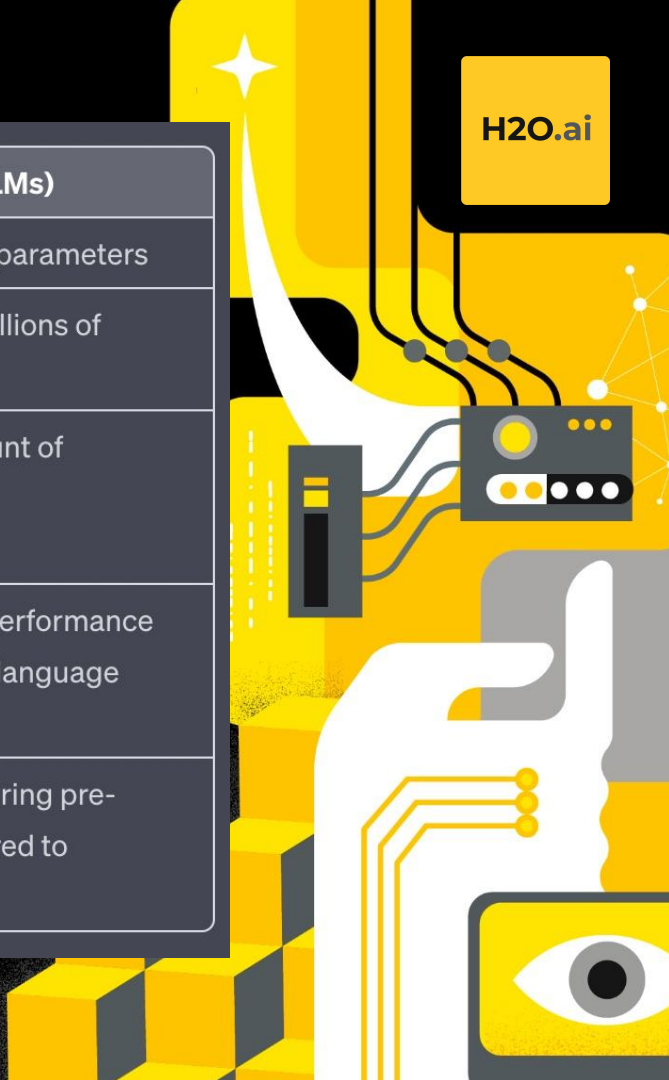
- The primary aim of Large Language Models is to acquire potent language representations from extensive text data.

- Once they have gained this expertise, they can undergo fine-tuning for specific language tasks.

- These tasks may include sentiment analysis, question-answering, or text classification, among others.

H2O.ai Confidential

Concept	Original Transformer	Large Language Models (LLMs)
Model Size	Relatively small	Enormous, with many more parameters
Training Data	Moderate-sized datasets	Vast datasets, sometimes billions of words
Training Computational Power	Requires less computing power	Demands a significant amount of computing power
Applicability	Excels in specific tasks like translation and sequence generation	Demonstrates remarkable performance across a broad spectrum of language tasks
Transfer of Knowledge	May not be as adaptable to a wide variety of language tasks	Acquires vast knowledge during pre-training, effectively transferred to numerous diverse tasks



Key Steps in Fine-Tuning	Summary
Task-specific Dataset	Collect or create a dataset with labeled examples tailored to the desired task.
Model Selection	Choose a pre-trained language model like BERT to leverage prior knowledge.
Training Objective	Define a clear goal for training, specifying what the model should learn.
Fine-Tuning Process	Feed the task-specific dataset into the pre-trained model for practice and adjustment.
Iterative Improvement	Model parameters are optimized over multiple rounds, enhancing task performance.

Model/Application	Description
h2ogpt	Provides fine-tuned models like LLaMa2 and llama.cpp for tasks such as chatting, document summarization, and tailored questions. Supports CPU/GPU, valuable for natural language processing across domains. Includes specialized models like finGPT and healthcareGPT.
BERT for Sentiment Analysis	BERT fine-tuned for sentiment analysis, determining positivity, negativity, or neutrality in sentences.
GPT-3 for Chatbots	GPT-3, a powerful language model, fine-tuned as a chatbot for human-like conversations in online applications.
RoBERTa for Named Entity Recognition (NER)	RoBERTa fine-tuned to recognize named entities like people, organizations, and locations in text.

Transfer learning

- Uses a pre-trained model as a foundation for a new task.
- Instead of starting from scratch, the model begins with pre-trained weights.
- Fine-tunes on a smaller labeled dataset specific to the new task.
- Adapts pre-learned representations to the new data's patterns and characteristics.
- Ideal for tasks with limited labeled data or resource-intensive training.

Adaptation (domain adaptation)

- Targets domain differences between source and target domains.
- Its goal is to make a model trained on the source domain perform well on the target domain, even with limited labeled data.
- A key challenge is ensuring effective generalization despite distribution shifts.
- Adaptation techniques align representations from the source domain with the target domain to reduce domain discrepancies.

Robot Adaptation Approaches

1. **Feature-based adaptation**: Simplifies the robot's view by finding common features between old and new objects.

2. **Instance-based adaptation**: Adjusts the robot's focus by prioritizing similar objects in the new environment.

3. **Model-based adaptation**: Fine-tunes the robot's recognition abilities by emphasizing relevant details in the new environment.

H2O.ai Confidential

- Fine-tuning in LLMs enhances adaptation.
- Empowers models with styles, personalities, and domain knowledge.
- Starts with a pre-trained LLM.
- Pre-training is generic, lacks specificity.

H2O.ai Confidential

- Knowing LLM architecture empowers researchers and practitioners.
- Enables capturing context, managing long-range connections, and producing quality results.
- Enhances application design, boosts model performance, and improves language-related tasks.

H2O.ai Confidential

H2O.ai University

Thank you!